

Zhirui Dai

+1 (858) 200 5325 - [Website](#) - [Google Scholar](#) - zhdai@ucsd.edu - [LinkedIn](#) - [GitHub](#)

EDUCATION

University of California San Diego

Ph.D. in Electrical and Computer Engineering

San Diego, CA, USA

01/2022 - Current (Expected 06/2027)

University of California San Diego

M.S. in Electrical and Computer Engineering

San Diego, CA, USA

09/2019 - 06/2021

Fudan University

B.S. in Physics

Shanghai, China

09/2015 - 06/2019

TECHNICAL SKILLS & OPEN SOURCE CONTRIBUTIONS

Programming Languages: C++, CUDA, Python, C, MATLAB, Java, Swift

Libraries and Tools: PyTorch, ROS, OpenCV, Open3D, Docker, Kubernetes, LangChain, ManiSkill

Made Open Source Contributions to: [PyTorch](#), [Open3D](#), [PointCloudLibrary](#), [pybind11](#)

RESEARCH

I am passionate about enabling autonomous robots to safely navigate and interact in complex environments. My research focuses on 3D scene representation learning, vision-language models for robotics, and robot task planning with large language models. I leverage CUDA, C++, Python etc. to develop algorithms that allow robots to perceive, reason, and act in the real world.

Research Assistant, UCSD Existential Robotics Lab

San Diego, CA, USA

Advisor: Prof. Nikolay Atanasov

01/2022 - Current

• Vision-Language Models for 3D Scene Understanding & Manipulation [RSS 2025, ICRA 2026]

- Architected [SBP](#), an incremental 3D language-feature mapping framework that provides **spatiotemporal memory** for mobile manipulators in unseen environments.
- Implemented **hierarchical hash-grid maps** with PyTorch to encode high-dimensional vision-language features **from 2D VLMs** (e.g., CLIP, DINO) into a lower-dimensional latent space and a **cross-scene generalized decoder** to reconstruct the original features.
- Developed a **transformer-based manipulation policy** conditioned on the 3D language-feature map, outperforming image-based policies in both in-distribution and novel scenes (up to **15% higher success rate**).
- Engineered data pipelines in **ManiSkill-HAB** to synthesize large-scale multi-modal datasets (RGB-D, semantic labels, task descriptions) for training the decoder and the manipulation policy.

• Neural Signed Directional Distance Functions (SDDF) for 3D Reconstruction [TPAMI 2026, RSS 2025 Workshop Best Paper]

- Introduced [SDDF](#), a novel 3D representation enabling single-pass computation for occlusion reasoning and collision checking.
- Designed a hybrid neural model combining ellipsoidal priors with neural residuals to learn scene-level SDDF from point clouds, achieving **SOTA reconstruction accuracy** on Replica dataset.
- Engineered custom [CUDA kernels](#) for ellipsoidal SDDF computation, indexed pose transformations and indexed linear layer, achieving up to **600x speedups** and **50-90% GPU RAM savings** compared to PyTorch implementations.
- Deployed the model on aerial platforms to demonstrate real-time sensor view coverage maximization and collision avoidance.

• Real-time Signed Distance Function (SDF) Mapping & Motion Planning [IROS 2026, RA-L 2026, IJRR 2025]

- Proposed [OREN](#), an octree residual network for globally accurate, real-time SDF learning from LiDAR/RGB-D, outperforming SOTA in accuracy and efficiency while running in real time (**14 fps** on desktop GPU, **7 fps** on Jetson Orin).
- Open-sourced the first [semi-sparse octree library](#) in C++ with **custom CUDA kernels**, accelerating voxel indexing by up to **550x** and Morton code computation by up to **13x**.
- Extended OREN to [OREN-X](#), jointly training SDF, occupancy, radiance and **vision-language fields** in a single octree at **30+ fps**, improving near-surface SDF accuracy by **33%** and open-vocabulary 3D mIoU by **71%** over prior methods.
- Developed [Kernel-SDF](#), an open-source C++ kernel regression library for real-time SDF estimation with **uncertainty quantification**, featuring full Python and ROS interfaces.
- Co-designed SDF mapping and [distance-accelerated motion planning](#) (Bubble*) for UAVs, running onboard a quadrotor and planning ≈ 90 m trajectories in **1-3 s** vs. up to 10 s for baselines.

• Large Language Model (LLM) for Robotic Task Planning [ICRA 2024, IROS 2025]

- Built an [optimal scene graph planner](#) utilizing **LangChain** to translate natural language into LTL specifications.
- Proposed a scene graph planning algorithm that leverages consistent LTL heuristics and LLM guidance to achieve **1600x speedup** over AMRA* while guaranteeing plan optimality.
- Proposed [LTLCodeGen](#), utilizing prompt engineering to generate **100% syntactically correct** LTL specifications, outperforming fine-tuned baselines (e.g., BART-large).

Research Assistant, UCSD Statistical Visual Computing Lab

San Diego, CA, USA

Advisor: Prof. Nuno Vasconcelos

06/2020 - 08/2021

• Social Distance Monitoring from Surveillance Videos [ICCV 2021]

- Developed [BEV-Net](#), a real-time social distance monitoring system (for COVID-19 pandemic) utilizing a novel **Bird's-Eye-View (BEV) projection layer** to map image features to BEV space.
- Implemented a calibration-free approach by recovering camera pose (height and pitch) via human keypoint detection and human heights.
- Achieved **30% lower error** in distance estimation and **2x higher IoU** for risk zone detection compared to SOTA methods.

PUBLICATIONS

Preprints

- Z. Dai, Q. Qian, D. M. Nguyen, Q.-D. Pham, K. Bronder, C. Nieto-Granda, Y. Chen, Q. Nguyen, N. Atanasov. “**OREN-X: Octree Residual Network for Real-Time Multi-Modal Mapping.**” Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027, under review. [Paper Project](#)
- J. Stanley*, Z. Dai*, Q. Qian, T.-C. Ho, T. Fan, S. Saha, C. Barngrover, K. M. B. Lee, N. Atanasov. “**From Distances to Trajectories: Real-Time Signed Distance Function Mapping and Distance-Accelerated Motion Planning for UAVs.**” Submitted to Autonomous Robots, 2026, under review. [Paper Project](#) (*Equal contribution)
- K. M. B. Lee, Z. Dai, C. Le Gentil, L. Wu, N. Atanasov, T. Vidal-Calleja. “**Safe Bubble Cover for Motion Planning on Distance Fields.**” arXiv preprint arXiv:2408.13377, 2024. [Paper](#)

Journal & Conference Papers

- Z. Dai*, T. Fan, M. Amani, J. Seo, K. M. B. Lee, H. Oh, N. Atanasov. “**Kernel-SDF: An Open-Source Library for Real-Time Signed Distance Function Estimation using Kernel Regression.**” IEEE Robotics and Automation Letters (RA-L), 2026. [Paper Project](#) (*Equal contribution)
- Z. Dai*, Q. Qian*, T. Fan, N. Atanasov. “**OREN: Octree Residual Network for Real-Time Euclidean Signed Distance Mapping.**” IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. [Paper Project](#) (*Equal contribution)
- T. Matthews, M. Kumar A. R., A. Begzadić, S. Tonkens, Z. Dai, N. Atanasov, J. Cortés, S. Herbert. “**Expanding Safe Sets with Learning-Based Barrier Functions and Reachability.**” IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. [Paper Project](#)
- Z. Dai, H. Shin, Y. Tian, K. M. B. Lee, N. Atanasov. “**Learning Scene-Level Signed Directional Distance Function with Ellipsoidal Priors and Neural Residuals.**” IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2026. [Paper Project](#)
- S. Kim, W. Chung, Z. Dai, D. Bhatt, A. Shukla, H. Su, Y. Tian, N. Atanasov. “**Seeing the Bigger Picture: 3D Latent Mapping for Mobile Manipulation Policy Learning.**” IEEE International Conference on Robotics and Automation (ICRA), 2026. [Paper Project](#)
- B. Rabiei, M. Kumar A. R., Z. Dai, S. L. S. R. Pilla, Q. Dong, N. Atanasov. “**LTLCodeGen: Code Generation of Syntactically Correct Temporal Logic for Robot Task Planning.**” IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025. [Paper Project](#)
- K. Long, Y. Yi, Z. Dai, S. Herbert, J. Cortés, N. Atanasov. “**Sensor-Based Distributionally Robust Control for Safe Robot Navigation in Dynamic Environments.**” The International Journal of Robotics Research (IJRR), 2025. [Paper Project](#)
- Z. Dai, A. Asgharivaskasi, T. Duong, S. Lin, M.-E. Tzes, G. J. Pappas, N. Atanasov. “**Optimal Scene Graph Planning with Large Language Model Guidance.**” IEEE International Conference on Robotics and Automation (ICRA), 2024. [Paper Project](#)
- Z. Dai, Y. Jiang, Y. Li, B. Liu, A. B. Chan, N. Vasconcelos. “**BEV-Net: Assessing Social Distancing Compliance by Joint People Localization and Geometric Reasoning.**” IEEE/CVF International Conference on Computer Vision (ICCV), 2021. [Paper](#)

Workshop Papers

- J. Stanley*, Z. Dai*, T.-C. Ho, S. Saha, C. Barngrover, N. Atanasov. “**From Distances to Trajectories: Real-Time Signed Distance Function Mapping and Distance-Accelerated Motion Planning for UAVs.**” ICRA 2026 Workshop on Aerial Inspection for Marine Infrastructures (AIMI), 2026. [Paper Project](#) (*Equal contribution)
- Z. Dai, H. Shin, Y. Tian, K. M. B. Lee, N. Atanasov. “**Learning Scene-Level Signed Directional Distance Function for Aerial Autonomy.**” Workshop on Leveraging Implicit Methods for Aerial Autonomy at RSS, 2025. [Paper Project](#) [Best Paper Award]
- S. Kim, W. Chung, Y. Tian, Z. Dai, A. Shukla, H. Su, N. Atanasov. “**Seeing the Bigger Picture: 3D Latent Mapping for Mobile Manipulation Policy Learning.**” Workshop on Mobile Manipulation and Workshop on Learned Robot Representations at RSS, 2025. [Paper Project](#) [Best Paper Nomination]

WORK EXPERIENCE

Research Internship — VR/AR Systems & Deep Learning

XCOM Labs Inc., CA, USA

08/2021 - 12/2021

- Engineered a neural pose prediction algorithm for multi-user VR/AR remote rendering systems, minimizing motion-to-photon latency.
- Designed and trained a neural network that achieves **90% lower error** in trajectory prediction compared to the default OpenVR baseline.
- Curated a proprietary 6-DoF dataset for XR head & hand pose forecasting to benchmark predictive models.

Software Development Internship — SoC Software Development

Camel Microelectronics Inc., CA, USA

01/2018 - 06/2019

- Architected **CamelStudioX**, a comprehensive IDE for low-power SoC software development for educational and commercial users. [\[Code\]](#)
- Developed embedded infrastructure: **Hardware Abstraction Layers (HAL)**, soft-float libraries, cross-compiler toolchains, ELF tools. [\[Code\]](#)
- Collaborated with hardware engineers on SoC validation, developing software patches as hardware hotfixes.

SERVICES

Teaching Assistant, [ECE276B: Planning & Learning in Robotics](#)

Spring 2023, Spring 2024

Mentor, [ENLACE](#), [SRIP](#), [STARS](#) Summer Research Programs

07/2022 - 08/2022

Mentor, [Guided Engineering Apprenticeship in Research \(GEAR\)](#)

09/2021 - 07/2022

AWARDS & HONORS

Electrical and Computer Engineering Department Fellowship

2022

Outstanding Graduate of Fudan University

2019